

Madhushree Mandokar

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madhu-portfolio-rho.vercel.app

Summary

Final year Computer Science Engineering student with hands-on proficiency across Generative AI, Machine Learning engineering, NLP pipelines, RESTful APIs, and Full-Stack Web Development. Expert at architecting production-ready, data-driven applications utilizing Python, Flask, React.js, and Vector Databases. Strong foundation in advanced Data Structures, Object-Oriented Programming (OOP), and analytical troubleshooting with a track record of implementing high-accuracy software solutions.

Technical Skills

- **Programming Languages:** Python, Java, SQL, JavaScript, C++, HTML, CSS
- **Frontend Development:** React.js, Responsive UI Design, Modern JavaScript (ES6)
- **Backend Development:** Flask Framework, RESTful API Engineering, Microservices
- **Machine Learning & AI:** NLP, TF-IDF Vectorization, Cosine Similarity, Scikit-learn, Classification Algorithms, Generative AI, RAG
- **Libraries & Tools:** Pandas, NumPy, Matplotlib, Seaborn, Git, GitHub Version Control, Streamlit, Vercel
- **Databases:** MySQL, MongoDB, Vector Databases (ChromaDB/Pinecone)
- **Core Concepts:** Data Structures and Algorithms (DSA), OOP, DBMS, Artificial Intelligence Fundamentals

Experience

BrainyBeam Info-Tech Pvt. Ltd.

May 2026 – Jun 2026

Data Science & Machine Learning Intern

- Managed 5+ end-to-end Data Science production pipelines, streamlining exploratory data analysis (EDA) and data validation tasks.
- Engineered, evaluated, and cross-validated 6 distinct predictive models including SVM, K-NN, Linear Regression, Decision Trees, and Random Forests in Python.
- Boosted predictive precision parameters by optimizing hyper-parameters against mathematical performance metrics, analyzing MAE, MSE, RMSE, and R^2 scores.
- Collaborated across a multi-functional technical team of developers to structure robust extraction workflows, resolving high class-imbalances across datasets of 10,000+ data instances.

Projects

- **LexAssist — RAG-Based Legal AI Assistant** github.com/purplemadhu2910/lexassist — Live
 - Engineered a Retrieval-Augmented Generation (RAG) system reducing hallucination rates to deliver contextual insights from documents.
 - Executed structured document text chunking, embedding vectors generation, semantic search optimization, and vector database indexing.
 - Implemented high-performance backend REST APIs leveraging FastAPI to handle document ingestion pipelines and prompt-response routing seamlessly.
 - Constructed a responsive frontend interface using Streamlit and React.js to support clean file uploads and multi-turn conversational queries.
- **AI Resume Screening System** github.com/purplemadhu2910/algorithm_visualizer — Live
 - Architected an automated processing tool matching candidate profiles to descriptions using advanced Natural Language Processing.
 - Applied token TF-IDF vectorized indexing models and cosine spatial comparisons to compute resume compatibility metrics.
 - Formulated a streamlined full-stack dashboard leveraging Flask, React.js, and SQL to accelerate processing speeds under low latency.
- **UPI Fraud Detection System** github.com/purplemadhu2910/upi_fraud_detection — Live
 - Modeled an anomaly identification platform parsing transaction streams to isolate illicit threats with an 80% accuracy threshold.
 - Conducted exploratory scaling transformations and binary target imbalance correction steps using Pandas, NumPy, and Scikit-learn.
 - Shipped model logic onto an interactive Streamlit dashboard rendering dynamic tracking charts and threat telemetry matrices.

Education

Parul University

Bachelor of Engineering in Computer Science

2023 – Present

CGPA: 7.87 / 10.00

Certifications

- Artificial Intelligence & Deep Learning Foundations — Infosys Springboard
- Natural Language Processing Specialization — Infosys Springboard
- Dynamic Programming Camp — AlgoUniversity
- SQL and Relational Databases — IBM SkillsBuild

Achievements

- Solved **200+** Data Structures and Algorithms problems across platforms including LeetCode.
- Competed in Vadodara Hackathon contributing codebase functionality within cross-functional team project sprints.
- Built and deployed multiple AI-powered applications using RAG, Machine Learning and NLP technologies.